# Xiangling Xu (许湘灵)

My Website
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Google Scholar
Orcid

**Research Interests**: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

## Education

2023 – now — **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*

2020 – 2023 — **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00

2016 – 2020 — **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Noncommutative Geometry in Gravity, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

## Publications and Preprints

1. M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. URL: https://arxiv.org/abs/2509.25145.
2. I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. URL: https://arxiv.org/abs/2507.17006.
3. **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. URL: https://arxiv.org/abs/2505.22309.
4. M.-O. Renou, **X. Xu**, and L. T. Ligthart, *Two convergent npa-like hierarchies for the quantum bilocal scenario*, 2024. arXiv: 2210.09065 [quant-ph]. URL: https://arxiv.org/abs/2210.09065.
5. X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. DOI: https://doi.org/10.1093/mnras/stac3500.
6. **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023. DOI: https://doi.org/10.3929/ethz-b-000613670.

## Conferences, Seminars, Workshops

### Contributed Talks

Oct 2025 — *YQIS25*, Institute of Photonic Sciences (ICFO)

## Conferences, Seminars, Workshops (continued)

| | | |
|---|---|---|
| Jul 2025 | | *IWOTA 2025 Twente*, University of Twente |
| Jun 2025 | | *IQC-PCQT-Quantum Saclay Workshop*, Sorbonne University |
| Sep 2024 | | *Causalworlds*, Perimeter Institute for Theoretical Physics |
| Nov 2023 | | *Workshop Defi EQIP*, Inria Lyon |
| Oct 2023 | | *Saclay Quantum Seminar*, Inria Saclay |

### Poster Sessions

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| Nov 2024 | | *Colloquium GDR TeQ*, Sorbonne University |
| | | *YQIS24*, Inria de Paris |

### Other Participation

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|---|---|---|
| Mar 2025 | | 828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef |
| Sep 2024 | | VQF Vienna Quantum Foundations Conference, IQOQI Vienna |
| Nov 2023 | | POP23 Workshop on Future Trends in Polynomial OPtimization, LAAS-CNRS Toulouse |

## Collaborations

### Group Visits

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|---|---|---|
| Oct-Nov 2025 | | *Thomas Vidick*, EPFL, Lausanne<br>Invited two-week visit |
| Sep 2024 | | *Igor Klep*, University of Ljubljana<br>Invited two-week visit funded by Erasmus+ Mobility program |
| Jul 2024 | | *William Slofstra*, Institute for Quantum Computing, University of Waterloo<br>Invited two-week visit |
| Dec 2023 | | *David Gross*, University of Cologne<br>Invited one-week visit |

### Organizations

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|---|---|---|
| Mar 2025 - Now | | Coorganizer for *Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory*<br>Coordinating the Gaspard Monge Visiting Professor program for Igor Klep<br>Co-planning the seminar and the schedule for invited speakers |
| Mar 2025 | | Coorganizer for *828. WE-Heraeus-Seminar at Bad Honnef*<br>Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts<br>Coordinating with the organizers and inviting some of the speakers |
| Feb 2024 - Now | | Coorganizer of *Project COMPUTE for Quantera*<br>Coordinating with the members<br>Making the project poster and website |
| Sep 2023 | | Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*<br>Planning and organizing the one-day program to bring all quantum groups at Saclay together |

## Awards and Scholarships

2024 *Erasmus+ Mobility Program*, funds for visiting the University of Ljubljana

2020 *McNab Undergraduate Scholarship*, University of Toronto

2017 - 2020 *Dean's List Scholar*, University of Toronto

2018 *The Class of 3T0 and Associates Scholarship in Mathematics and Physics*, University of Toronto

*International Exchange Award*, University of Toronto

*Bronze Medal*, 2017 University Physics Competition

2017 *Eileen Doman Award*, University of Toronto

*Summer Undergraduate Research Program*, Canadian Institute for Theoretical Astrophysics

2016 *President's Scholar of Excellence Program*, University of Toronto

## Teaching Experience

### Lecture

Nov 2024 *Recent trends in parallel, distributed, and quantum computing*
Master 2 QDCS, Université Paris-Saclay

### Teaching Assistance

2022-2023 *Linear Algebra I & II*
Department of Mathematics, ETH Zürich

2021-2022 *Analysis I & II*
Department of Mathematics, ETH Zürich

## Skills

Languages Chinese (native), English (fluent), German (beginner)

Coding Python (Ncpol2sdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, LaTeX

## Referees

Available on Request